

To build your own full-stack AI Video Generator platform (a custom web application that takes user text/image prompts and returns generated videos), you need to master three main pillars: **AI Models**, **Full-Stack Web Development**, and **Cloud GPU / Asynchronous Infrastructure**.

Here is a complete, structured roadmap with direct English course links and documentation resources for each phase.

Phase 1: AI Video Generation Models & Pipelines

In this phase, you learn how text-to-video and image-to-video models work programmatically using Python and deep learning libraries.

- **Hugging Face Diffusers Library & Course**
 - - **Link:** [Hugging Face Diffusion Models Course](#) | [Diffusers Video Pipelines Documentation](#)
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 - **What you'll learn:** How to load open-source video models (e.g., Stable Video Diffusion, CogVideoX, AnimateDiff), pass text prompts, tune parameters (frame rate, motion scale, guidance steps), and export MP4 videos.
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- **Practical Deep Learning for Coders (Fast.ai)**
 - - **Link:** [Fast.ai Course](#)
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 - **What you'll learn:** Fundamentals of PyTorch, neural networks, and Generative AI required to customize or fine-tune models.
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- **ComfyUI API Execution**
 - - **Link:** [ComfyUI Documentation](#) | [RunPod ComfyUI API Guide](#)
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 - **What you'll learn:** How to turn visual node-based video generation workflows into automated API endpoints.
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Phase 2: Full-Stack Web Development (Frontend & Backend)

Your users need a web application to log in, type prompts, manage credits, and view generated videos.

- **FastAPI Backend Development**
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- **Link:** [FastAPI Official Documentation & Tutorial](#)
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- **What you'll learn:** Building high-performance Python APIs to handle video generation requests, handle WebSockets/SSE for real-time video generation progress, and authenticate users.
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- **Next.js & React Frontend Framework**
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- **Link:** [Next.js Learn Course](#)
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- **What you'll learn:** Building modern user interfaces, dashboard controls, prompt textboxes, media galleries, and handling real-time state.
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- **freeCodeCamp: Full-Stack AI Application Development**
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- **Link:** [freeCodeCamp Full-Stack AI SaaS Tutorial \(YouTube\)](#)
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- **What you'll learn:** Connecting Next.js frontends to Python backends, adding Stripe payments, and setting up Supabase/PostgreSQL databases.
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Phase 3: Cloud GPU Hosting & Asynchronous Queuing

Because video generation takes 20 seconds to several minutes, running models on a normal web server will crash it. You need Serverless GPUs and job queues.

- **Serverless GPU Hosting (Replicate / RunPod / Modal)**
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- **Link:** [Replicate API Documentation](#)
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- **Link:** [RunPod Serverless GPU Docs](#)
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- **Link:** [Modal Labs Docs](#)
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- **What you'll learn:** Deploying GPU-heavy AI video models to cloud servers that automatically scale up when users generate videos and scale down to zero when idle.
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- **Asynchronous Task Queuing (Celery + Redis)**
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- **Link:** [Celery First Steps with Python](#)
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- **What you'll learn:** Managing background tasks (e.g., *User clicks Generate* $\rightarrow$ *Task sent to queue* $\rightarrow$ *GPU processes video* $\rightarrow$ *Video saved to cloud storage* $\rightarrow$ *User notified*).
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- **Cloud File Storage (AWS S3 / Cloudinary)**
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- **Link:** [AWS S3 Python SDK \(Boto3\) Getting Started](#)
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- **What you'll learn:** Saving rendered `.mp4` video files to cloud bucket storage and delivering them fast via CDN.
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Recommended System Architecture Flow

To visualize how your full custom application will work:

